



Filmon Misgane

Applied Mathematician

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Citizenship: Norwegian



Profile

MSc candidate in Mathematics for Applications. Completing my degree this summer with a specialization in numerical PDEs and scientific computing. My experience includes Python-based solver development, FEM methods and spline methods. I am particularly interested in scientific software, numerical simulation and applied computational science.



Education

Present
↑
08.2024

MSc. Mathematics for Applications

University of Oslo (UiO)

- Specialized in Partial Differential Equations (PDEs).
- Master thesis: Numerical Methods for Nonlocal Conservation Laws.
- Designed and implemented Python solvers for first and second order numerical schemes for nonlocal conservation laws.

06.2024
↑
08.2021

BSc. Mathematics and Computer Science

University of Oslo (UiO)



Work Experience

Present
↑
04.2026

Summer Researcher

UiO Department of Geosciences in collaboration with SINTEF

- Working on scientific machine learning methods for subsurface energy storage, with a focus on Physics-Informed Neural Networks (PINNs) and Neural Operators.
- Contributing to the development of machine learning models for PDE-based subsurface flow and transport problems.

Present
↑
04.2026

Summer Intern

DSB group, Department of Informatics

- Working on a unified FEM inversion framework for time-harmonic shear-wave elastography.
- Collaborating on quantitative evaluation of reconstruction methods and software development for medical imaging applications.

Present
↑
08.2022

Mentor

ENT3R, University of Oslo (UiO)

- Motivate high school students to study STEM subjects while creating engaging activities and assisting with their homework.



GitHub

<https://github.com/Filmon492>



Skills

- Programming: Python, C++, MATLAB
- Scientific computing: Numerical methods for PDEs, FEM and spline methods
- Tools/Libraries: Git/GitHub, PyTest, FEniCSx, FreeFEM, NumPy, SymPy and Pandas



Job References

Professor Ulrik Skre Fjordholm
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Relationship: Master's thesis supervisor



Languages

Norwegian	Fluent
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English	Fluent
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